Mikko Liivak

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Education

Cornell University — Ithaca, NY
Intended M.Eng. Computer Science — *Expected Dec 2027*

Cornell University — Ithaca, NY
B.S. Computer Science, College of Engineering — *Expected May 2027*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java), Systems (C/C#), Systems Programming (Rust), Discrete Math, Machine Learning, Database Systems, Analysis of Algorithms
- Activities: Association of Computer Science Undergraduates, Cornell Outdoor Education Climbing Wall, Thread Fashion Magazine Photographer

Professional Experience

OneStream Software — New York, NY (Remote)
Software Engineering Intern — *Jun 2026 – Aug 2026*

- Owned end-to-end delivery of **3+** customer-facing **C#/.NET** features for enterprise financial reporting.
- Shipped configuration features for enterprise financial reporting software used by **1800+** global customers.
- Authored **100+** regression test cases in C# to validate features, ensuring code quality before each release.
- Collaborated across **4+** engineering teams to streamline deployment pipelines and accelerate release cadence.

PARES AI (Y Combinator S25) — San Francisco, CA
Software Engineer (Founding) — *Jun 2025 – Sep 2025*

- Deployed core features for an AI-driven real estate platform consolidating agent tools across **5+** firms.
- Developed **Node.js**/**TypeScript** backend for property ingestion and auth, reducing response times by **35%**.
- Leveraged **Claude API** for structured property summaries and metadata validation, cutting manual work by **60%**.
- Refactored **PostgreSQL** schemas and migrations, improving query performance by **40%** and supporting **3x** growth.
- Implemented RBAC for **20K+** records with role permission audits to ensure zero cross-tenant data exposure.
- Wrote database and endpoint unit tests using **Docker** and **GitHub Actions** to validate continuous deployment.

Cornell University — Ithaca, NY
Student Researcher, Datacenter Acoustics Study — *Jan 2026 – Present*

- Engineered a **Python** web scraper to extract **1000+** testimonies from news sources on datacenter noise impacts.
- Deployed a **Flask** web app on **Google Cloud Platform** to visualize **1000+** scraped data points in real time.
- Integrated OpenStreetMap and Overpass API to render an interactive map of **100+** US datacenter sites.

Cornell University — Ithaca, NY
Research Assistant to Prof. Christopher Csikszentmihalyi — *May 2025 – May 2026*

- Developed a **Python** data pipeline to ingest, validate, and store **10K+** lake environmental sensor readings.
- Upgraded a water drone's Pixhawk flight controller to collect lake sensor data with **50%** faster processing.
- Automated data cleaning, anomaly detection, and schema validation workflows, reducing manual effort by **75%**.

Technical Projects

Piste (Swift, SwiftUI, MapKit)

- Developed an iOS app in **Swift**/**SwiftUI** with **MapKit** to pin and annotate terrain features across ski resort maps.
- Integrated **CloudKit** for real-time annotation sync, enabling community terrain discovery across resorts.
- Crafted a dual-mode interface for personal ride journals and shared terrain maps, beta tested with **3** users.

3-Band Audio Equalizer Plugin (C++, JUCE)

- Architected a real-time 3-band equalizer plugin in **JUCE** for audio processing, achieving under **5ms** latency.
- Optimized **C++ DSP** pipelines with efficient buffer management, cutting CPU overhead by **30%** during playback.
- Designed interactive filter UI with real-time parameter automation and live frequency-domain visualization.

SBC Trainer Database App (OCaml)

- Led a 3-person team to build and test backend systems in **OCaml**, ensuring reliability via **50+** unit tests.
- Spearheaded architecture of an interactive SBC solver supporting **1000+** player combinations in EA Sports FC.
- Implemented search-based optimization algorithms to solve complex squad-building puzzles in under **10 seconds**.

Skills

Languages: Python, Java, TypeScript, JavaScript, C++, C, C#, Rust, SQL, OCaml, Swift, HTML/CSS, Assembly
Frameworks & Libraries: React, Express, Node.js, FastAPI, Flask, PyTorch, JUCE, .NET, SwiftUI, MapKit
Databases & Tools: PostgreSQL, Redis, Supabase, Docker, Git, CloudKit, Google Cloud Platform (GCP)
Backend & Infrastructure: REST APIs, API Design, CI/CD, Unit Testing, Claude API